

```
#include<stdio.h>
```

```
#include <stdlib.h>
```

```
struct Node {
```

```
    int data;
```

```
    struct Node* next;
```

```
};
```

```
struct Node* createNode(int data) {
```

```
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
```

```
    newNode->data = data;
```

```
    newNode->next = NULL;
```

```
    return newNode;
```

```
}
```

```
void insertEnd(struct Node** head, int data) {
```

```
    struct Node* newNode = createNode(data);
```

```
    if (*head == NULL) {
```

```
        *head = newNode;
```

```
        return;
```

```
    }
```

```
    struct Node* temp = *head;
```

```
    while (temp->next != NULL) {
```

```
        temp = temp->next;
```

```
    }
```

```
    temp->next = newNode;
```

```
}
```

```
void display(struct Node* head) {
```

```
    while (head) {
```

```
        printf("%d -> ", head->data);
```

```

        head = head->next;
    }
    printf("NULL\n");
}

```

```

void sortList(struct Node* head) {
    struct Node *i, *j;
    int temp;

    for (i = head; i != NULL; i = i->next) {
        for (j = i->next; j != NULL; j = j->next) {
            if (i->data > j->data) {
                temp = i->data;
                i->data = j->data;
                j->data = temp;
            }
        }
    }
}

```

```

void reverseList(struct Node** head) {
    struct Node *prev = NULL, *current = *head, *next = NULL;

    while (current != NULL) {
        next = current->next;
        current->next = prev;
        prev = current;
        current = next;
    }
    *head = prev;
}

```

```
void concatenate(struct Node** head1, struct Node* head2) {  
    if (*head1 == NULL) {  
        *head1 = head2;  
        return;  
    }
```

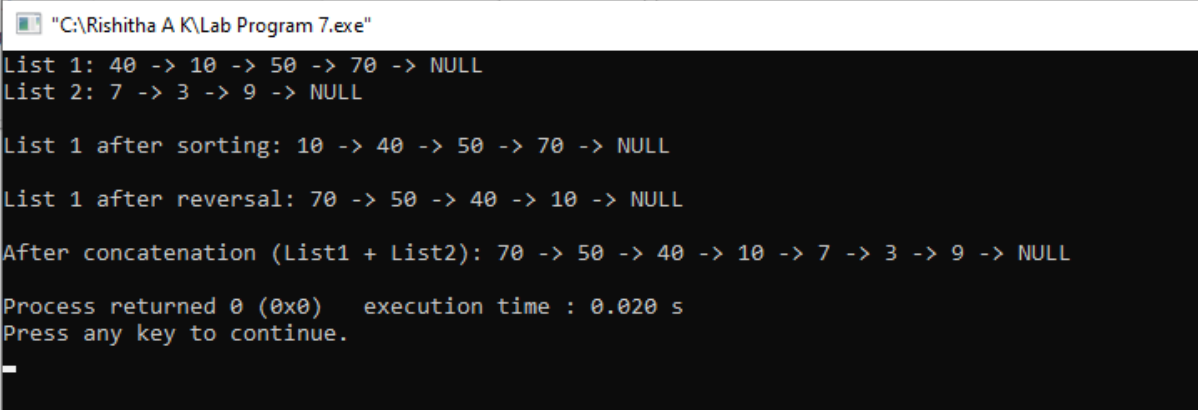
```
  
    struct Node* temp = *head1;  
    while (temp->next != NULL)  
        temp = temp->next;  
  
    temp->next = head2;  
}
```

```
int main() {  
    struct Node* list1 = NULL;  
    struct Node* list2 = NULL;  
    insertEnd(&list1, 40);  
    insertEnd(&list1, 10);  
    insertEnd(&list1, 50);  
    insertEnd(&list1, 70);
```

```
  
    printf("List 1: ");  
    display(list1);  
    insertEnd(&list2, 7);  
    insertEnd(&list2, 3);  
    insertEnd(&list2, 9);
```

```
  
    printf("List 2: ");  
    display(list2);  
    sortList(list1);
```

```
printf("\nList 1 after sorting: ");  
display(list1);  
reverseList(&list1);  
printf("\nList 1 after reversal: ");  
display(list1);  
concatenate(&list1, list2);  
printf("\nAfter concatenation (List1 + List2): ");  
display(list1);  
  
return 0;  
}
```



```
"C:\Rishitha A K\Lab Program 7.exe"  
List 1: 40 -> 10 -> 50 -> 70 -> NULL  
List 2: 7 -> 3 -> 9 -> NULL  
  
List 1 after sorting: 10 -> 40 -> 50 -> 70 -> NULL  
:  
List 1 after reversal: 70 -> 50 -> 40 -> 10 -> NULL  
  
After concatenation (List1 + List2): 70 -> 50 -> 40 -> 10 -> 7 -> 3 -> 9 -> NULL  
  
Process returned 0 (0x0)   execution time : 0.020 s  
Press any key to continue.  
_
```